

Article

The Influence Mechanism of Online Social Network Relationships on Sustainable Entrepreneurial Success

Xianhuan Chen ¹, Yiming Ma ^{2,*}  and Yuguang Xie ¹ 

¹ School of Management, Hefei University of Technology, Hefei 230009, China; chenxh216@hfut.edu.cn (X.C.); 13095537911@163.com (Y.X.)

² School of Management Science and Engineering, Anhui University of Finance and Economics, Bengbu 233030, China

* Correspondence: yimingma923@163.com

Abstract: Affected by economic uncertainty, the success rate of sustainable entrepreneurship is not optimistic at present. Effectively improving the success rate of sustainable entrepreneurship is of great significance for ensuring and maintaining sustainable development for long-term socioeconomic and environmental sustainability. Based on social network theory, entrepreneurial capital theory, psychological capital theory, and the literature on sustainable entrepreneurship, this study explores the influence mechanism of online social network relationships on sustainable entrepreneurial success from the perspective of those relationships, dividing entrepreneurs' abilities into an online relationship-building ability and online relationship coordination ability. We used an empirical study with 346 valid questionnaires. The results of a PLS-SEM analysis show that online social network relationships have a significant impact on sustainable entrepreneurial capital, in which online relationship-building capacity has a similar impact on both sustainable entrepreneurial resource integration and acquisition; online relationship-building capacity can positively affect both transactional and interpersonal psychological capital; sustainable entrepreneurial resource integration can be more effective than resource acquisition in promoting sustainable entrepreneurial success through the use of social media; and transactional psychological capital has a more significant impact on sustainable entrepreneurial success than does interpersonal psychological capital. This study provides new insights into how entrepreneurs can fully utilize their online social network relationships to achieve entrepreneurial success.

Keywords: online social network relationships; online relationship-building ability; online relationship coordination ability; sustainable entrepreneurial capital; psychological capital; sustainable entrepreneurial success



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1. Introduction

With the rapid changes in the international situation, the uncertain economic environment is becoming an important factor threatening entrepreneurs. Over the past 20 years, higher geopolitical risk has predicted lower investment and employment and has been strongly associated with higher-downside economic risks and failed bets [1,2]. At the same time, due to exchange rate changes and supply chain disruptions caused by turbulence in the international market, the degree of enterprise diversification in different regions has declined [3]. Additionally, most investors are more conservative in their investment behavior, often preferring stable or short-term business models. As a result, sustainable entrepreneurship oriented toward long-term interests and regional equity has become a non-priority option, and the difficulty of obtaining entrepreneurial resources has increased significantly. How to achieve the success of sustainable entrepreneurship in the current environment of economic uncertainty has become a key issue that entrepreneurs interested in sustainable development need to solve.

Sustainable entrepreneurship is a social endeavor in which individuals interact with others to help start, grow, and sustain emerging businesses [4]. Studies addressing sustainable entrepreneurial success have shown that achieving this success requires not only the support of factors such as markets, policies, and technologies but also entrepreneurial resources, and the amount of such resources available to an entrepreneur often plays a decisive role [5]. Because this amount is often insufficient to meet the sustainable entrepreneurship needs, entrepreneurs spend a lot of time and energy on their social networks searching for additional support [6]. A large body of research and practice confirms that entrepreneurial social network relationships facilitate many entrepreneurship-related outcomes, including opportunity discovery, access to scarce resources, changing business situations, and sustained business growth [7–9].

Over the past decade, with its rising popularity, social media has become a part of people's lives, and traditional offline social activities are rapidly shifting to online [10]. The current social media not only covers people's offline social networks but also expands the boundaries of social networking, making online social networks a part of the social networks of most people (including the elderly and children) [11]. Through social media, people can not only communicate with members of their social network but also acquire knowledge, information, and skills to accomplish tasks that were previously impossible to achieve. More and more entrepreneurs have sought to access entrepreneurship-related resources through online social networks to achieve start-up success [12]. Because the core of social media is social relationships, entrepreneurs use social media to convert social relationships into entrepreneurial capital [13]. Through social media, entrepreneurs can expand the size and quality of their social networks to more rapidly identify entrepreneurial opportunities, explore potential entrepreneurial partners, and establish deep partnerships [12]. In food production, for example, social media can help entrepreneurs access information resources, collaborate better with partners, discover new operational resources, and re-configure processes [14]. Additionally, some entrepreneurs are using social media beyond marketing to network for business, search for information, and obtain startup funding [15]. Effective communication based on social media can improve the organizational atmosphere and improve the efficiency of communication, so that entrepreneurs can obtain more psychological capital to support entrepreneurship [16]. This improves their access to entrepreneurial resources and increases entrepreneurial efficiency and the likelihood of success. Especially in the current economic uncertainty, the traditional financing and entrepreneurial resource model based on offline social networks makes it difficult to meet the entrepreneurial needs of sustainable entrepreneurship. Entrepreneurs need to find a larger pool of individuals or organizations with like-minded investors or who are willing to help with sustainable entrepreneurship. This further expands the important role of online social networks for sustainable entrepreneurship.

Obtaining relevant sustainable entrepreneurial resources from online social networks is not an easy task. From a competency perspective, an entrepreneur's online social network relationships represent an ability to succeed or not (i.e., they represent the person's relationship-building and relationship coordination abilities) [17]. Through these relationships (or capabilities), entrepreneurs can obtain entrepreneurial resources. However, there are different contents and types of entrepreneurial capital and different capital required by different firms. Especially for entrepreneurs who are interested in implementing sustainable entrepreneurship, in an uncertain economic environment, they need not only tangibly venture capital, but also intangible psychological capital to provide support for their entrepreneurship [18]. Therefore, further clarification is needed on how online social network relationships affect access to sustainable entrepreneurial capital. Additionally, given the multiplicity of influences (internal and external) generated by online social network relationships, a need exists to adopt a more precise delineation to explore the impact of these influences on entrepreneurial resources [19]. Based on this, this study asks the following questions to explore the impact of entrepreneurs' online social network relationships on their entrepreneurial success.

- (1) How do different types of online social network relational ability affect sustainable entrepreneurial capital?
- (2) How do different types of online social network sustainable relational ability affect sustainable psychological capital?
- (3) How do sustainable entrepreneurial and psychological capital affect entrepreneurial success?

To address these three questions, our work integrates past research and constructs a research model based on social network theory, entrepreneurial capital theory, and psychological capital theory to explore the mechanisms by which online social network relationships affect sustainable entrepreneurial success. Among these, social networks refer to the relationship networks formed by the interaction of individuals or families with their relatives, friends, colleagues, neighbors, and so on [20], which is crucial for entrepreneurs when starting a business. With the continuous development of internet technology, the connotation of individual social networks has gradually shifted from offline to online, and the connotation of entrepreneurs' social networks has also changed. In addition, entrepreneurial capital and psychological capital reflect entrepreneurs' ability to utilize resources and a positive psychological state during the entrepreneurial process, and these two factors are significant predictors of entrepreneurial success. Therefore, social network theory, entrepreneurial capital theory, and psychological capital theory also apply to this paper to explain the mechanism of entrepreneurs' social networks on sustainable entrepreneurial success. The goals are to provide new empirical evidence for entrepreneurs in a climate of economic uncertainty to help them better use online social networks, tap more sustainable entrepreneurial resources, and ultimately, sustainable entrepreneurial success in their endeavors.

2. Literature Review

2.1. Online Social Networks

A social network is a network of relationships formed by the interactions of an individual or family with their relatives, friends, colleagues, neighbors, and so on [20], and this network is also the specific structure of social relationships and patterns of connections possessed by the subjects of the network. At the heart of social networks are relationships, from which individuals can derive a range of social resources and social support through interpersonal communication and interaction [21]. With the continuous development of internet technology, online social networking software programs, such as WeChat, QQ, Facebook, Twitter, Sina Weibo, and so on, have become an indispensable part of people's lives, gradually shifting the connotation of an individual's social network from offline to online. The connotation of traditional social networks mainly refers to the networks formed by individuals or families interacting with their relatives, friends, colleagues, neighbors, and so on through offline channels [20]. By contrast, online social networks emphasize the relationship networks formed by individuals with others through online channels and online social networking software in the new media environment. Compared to traditional social networks, online social networks are more prosperous and have a complex division of dimensions.

The core of social network theory is relationships [17], and online social network relationships encompass the nature, strength, and quality of relationships between individuals and members of their social network, which include both online social trust and online social relationships. From the perspective of resource access, online social relationships can be categorized into relationships intended to build one's network and relationships to increase one's coordination ability. Network building refers to constructing a personal network of relationships among the subjects of a social network. The earliest examples were managers actively engaging in activities in their social networks to build personal networks of relationships [22]. This network building, in turn, was defined as a set of organized behaviors to build and maintain informal relationships with others [23]. Online coordination ability refers to selecting, absorbing, exchanging, allocating, activating, and integrating different types of resources. It is the subjective ability to identify, acquire,

allocate, and utilize resources from social network relationships in the entrepreneurial process [24].

With the continuous development of social network theory in entrepreneurship research, social networks have become an essential perspective in explaining various issues in the entrepreneurial process [25]. Previous studies have pointed out that social networks play an important role in the entrepreneurial process. Studies have shown that social networks are important for entrepreneurial self-efficacy [26], entrepreneurial enthusiasm [27], entrepreneurial willingness [21], and entrepreneurial learning [28]; innovative firm performance [29], organizational performance [30], and the performance growth of SMEs [31] have significant impacts that are validated by data analysis. Similarly, in the internet era, establishing online (versus offline) social networks is more in line with current trends. Online networks are more capable of providing individuals with rich social resources and social support, including tool resources, information resources, emotional resources, and so on [32], and helping them achieve their goals [33]. Therefore, this study focuses on the impact of online social network relationships on entrepreneurship from the perspective of those networks by categorizing entrepreneurs' capabilities into an online relationship-building ability and online relationship coordination ability.

2.2. Sustainable Entrepreneurial Success

"Success" refers to the achievement of some measure that has been previously envisioned. The concept of entrepreneurial success, which extends from the concept of professional success, is an important indicator of entrepreneurial outcomes. No consensus exists on the definition of entrepreneurial success due to differences in research perspectives. For example, Lipper states that entrepreneurial success is the entrepreneur's assessment of his or her values and goals [34]. Subsequently, Lechler states that entrepreneurial success is the successful construction of a business through the power of social networks [35]. Additionally, Caliendo and Kritikos state that entrepreneurial success is the sustainable operation of a business [36]. However, limiting entrepreneurial success to economic indicators does not reflect its whole meaning; many entrepreneurs do not equate success with wealth, and a variety of psychosocial factors, such as achieving a feeling of self-worth, work-life balance, prestige, and/or recognition, influence entrepreneurs' perceptions of their own success [37].

Although different scholars define entrepreneurial success differently, a consensus exists on the notion that entrepreneurial success is a complex phenomenon that does not have a single dimension and consists of multiple indicators, both financial and non-financial. The financial indicators are operating margins, cash flow, total operating turnover, business growth rate, market sales, return on investment, and operating profit margin. Non-financial indicators include technological innovation, the growth rate of the number of employees, the number and satisfaction of customers, the number of new products developed, the image and reputation of the company, and the usefulness of products and services [38].

For sustainable entrepreneurship, the goal is to "preserve nature, sustain life, and communities pursue perceived opportunities to realize future products, processes, and services for gain, which is broadly interpreted to include economic and non-economic gains for individuals, the economy, and society" [39]. Therefore, the assessment of the success of sustainable entrepreneurship needs to consider not only the current economic benefits but also the future social and environmental impacts. As Sarah, Alvaro, and Leandro (2022) pointed out, philanthropic CSR is an important factor in the evaluation of the success of sustainable entrepreneurship and can reflect the long-term social and environmental benefits of economic development [40]. In another study, finding ways to monetize future products, nature conservation, life support, and communities is a key factor for entrepreneurs to implement sustainable entrepreneurship and an important indicator of sustainable entrepreneurial success [41]. In addition, in the study by He et al. (2020), reducing resource consumption and continuous institutional innovation are also effective measures to achieve sustainable entrepreneurial success [42]. Therefore, this

study believes that the success of sustainable entrepreneurship is not only about obtaining economic benefits, but more importantly, it is necessary to combine multiple factors, such as environmental, social, and economic value creation, to ensure the well-being of future generations and achieve sustainable and long-term benefits [43].

Many factors influence the realization of sustainable entrepreneurial success. The current factors can be sorted into intrinsic personality traits and extrinsic environmental factors. Personality traits include entrepreneurial age [44], opportunity recognition [45], self-efficacy [40,46], human capital [18,47], and entrepreneurial perseverance [48,49]. Environmental factors include entrepreneurship education [50], social resources [51], and family support [52,53]. In other words, a combination of an individual's entrepreneurial and psychological capital determines their sustainable entrepreneurial success. This study, therefore, explores these two aspects of capital for their impact on sustainable entrepreneurial success.

3. Model Construction and Hypothesis Development

3.1. Online Social Network Relationships and Sustainable Entrepreneurial Capital

Each person has a unique life background that creates the different resources available to them, both their own and those of people they know. The same is true for sustainable entrepreneurship. In an increasingly competitive environment, firms can achieve sustainable growth only by identifying, acquiring, constructing, reconfiguring, and generating internal and external entrepreneurial resources [54]. In this, the entrepreneur's social network plays an important role. However, compared with traditional offline social networks, online social networks have stronger scalability and can help entrepreneurs obtain more entrepreneurial resources. Especially in the current economic uncertainty, limited by the regional economic forum, entrepreneurs with sustainable entrepreneurial intentions often find it difficult to obtain sufficient entrepreneurial resources from offline social networks. Additionally, online social networks based on social media can find more sustainable entrepreneurship resource providers with common aspirations or investors interested in sustainable entrepreneurship. This helps to strengthen the entrepreneurial capacity and determination of sustainable entrepreneurs [55]. Connectivity theory in the age of social media suggests that entrepreneurs can access different sustainable entrepreneurial resources through social networks. Different ways to use resources exist; entrepreneurs can access relevant resources through social networks to power the development of their businesses, or they can form entrepreneurial resources with unique characteristics by optimally combining social relationships in their social networks [56]. That is, accessing resources requires building and coordinating online relationships.

The relationship-building ability of social networks is the ability to help individuals connect with others. In the era of social media, entrepreneurs' online relationship building is a kind of relationship-building ability based on social media [57]. Based on this ability, entrepreneurs are able to establish mutually beneficial long-term relationships with other members of the social network, conduct business learning, construct appropriate business plans, etc. Entrepreneurs with strong online relationship-building abilities can be more agile in searching for the information resources they need. Through their relationships, they can access the resources required for entrepreneurship (such as human, financial, and customer resources) from more channels [58]. They can also narrow the knowledge gap between different individuals [59], which in turn, enables access to entrepreneurial resources and achieving sustainable entrepreneurship [60].

For entrepreneurs, obtaining start-up resources from social networks is not the same as being able to use those resources effectively. Effective resource integration and portfolio is often the key to entrepreneurial success [61]. For example, research on entrepreneurs in family businesses shows that entrepreneurs can obtain entrepreneurial resources from family social networks [62]. However, if there is no effective resource integration, it is often difficult to achieve entrepreneurial success. Homogeneous entrepreneurial resources do not necessarily bring competitive advantage. Only the combination of different types

of resources to create resources with scarcity is often the key to entrepreneurial success. In other words, the integration of entrepreneurial resources is the coordination of social network relations. Online network relationship coordination ability is defined as an individual's ability to integrate resources based on social network relationships on social media. This capacity is especially important for sustainable entrepreneurship. For example, effective links between upstream and downstream supply chains; integrated supply among multiple suppliers; and benefit synergy of different retail terminals [63,64]. The solution of these problems requires the relationship coordination ability of sustainable entrepreneurs to effectively achieve successful sustainable entrepreneurship in the current economic environment.

Therefore, based on the above analysis, this study proposes the following hypothesis:

H1a. *Online relationship-building ability is positively related to sustainable entrepreneurial resource integration.*

H1b. *Online relationship-building ability is positively related to sustainable entrepreneurial resource acquisition.*

Entrepreneurs with strong online relationship coordination abilities can use their online social network relationships more efficiently to form the personal capital needed for sustainable entrepreneurship (i.e., financial, human, and social capital [65]) by integrating those relationships, which in turn leads to the acquisition of a variety of resources needed for sustainable entrepreneurship [66]. Such entrepreneurs are not only able to identify potential entrepreneurial capital (such as information, human, technology, and financial capital) in their networks, but they are also able to reasonably assess, configure, and use these resources; they interact with others and co-create value to maximize the effectiveness of the resources and effectively integrate them into the business [54]. When entrepreneurs can exchange and integrate entrepreneurial resources this way, they mobilize resources efficiently and optimize the value of co-creation outcomes among network participants at a given time to co-create social and economic value [67]. Therefore, based on the above analysis, this study proposes the following hypothesis:

H2a. *Online relationship coordination ability is positively related to sustainable entrepreneurial resource integration.*

H2b. *Online relationship coordination ability is positively related to sustainable entrepreneurial resource acquisition.*

3.2. Online Social Network Relationships and Psychological Capital

Psychological capital includes all the psychological resources that can promote the smooth running of entrepreneurial activities. For the entrepreneurs of sustainable entrepreneurship, psychological capital often plays a more important role because of the particularity of entrepreneurial mode and purpose [18]. To better analyze the impact of venture capital on Chinese entrepreneurs' sustainable entrepreneurship, this study further subdivides psychological capital. It is further categorized into interpersonal psychological capital and transactional psychological capital according to the Native Psychological Capital Scale. In terms of interpersonal psychological capital, when entrepreneurs have strong online relationship-building and relationship coordination abilities, they can build good interpersonal networks and integrate and use them, contributing to their ability to have good interpersonal interactions and show a positive mindset during such interactions [68]. Transactional psychological capital, however, refers to an important psychological quality for overcoming difficulties during the entrepreneurial process; having strong online relationship-building and relationship coordination skills enables entrepreneurs to accumulate appropriate "relational resources" from their social interactions [69]. Transactional

psychological capital provides entrepreneurs with a strong online network of relationships and the accompanying resources and emotional support, and the greater the perceived support, the more entrepreneurs can display positive attitudes, persevere, and succeed [70]. Therefore, based on the above analysis, this study proposes the following hypotheses:

H3a. *Online relationship-building ability is positively related to interpersonal psychological capital.*

H3b. *Online relationship-building ability is positively related to transactional psychological capital.*

H4a. *Online relationship coordination ability is positively related to interpersonal psychological capital.*

H4b. *Online relationship coordination ability is positively related to transactional psychological capital.*

3.3. Sustainable Entrepreneurial Capital and Sustainable Entrepreneurial Success

Firms' competitive advantage derives from their unique resources and how these resources are allocated. The essence of capability is identifying and integrating resources to ultimately improve performance [71]. Resource integration can be divided into three stages, including construction, clarification, and utilization. Moreover, resource integration at each stage has an irreplaceable role in enhancing enterprise performance [72]. An entrepreneur's ability to integrate resources leads to their firm's performance, and good performance allows them to further integrate existing resources into scarce or unique corporate resources, improving the firm's competitive advantage and thus increasing the likelihood of success. Therefore, we propose the following hypotheses:

H5a. *Entrepreneurial resource integration is positively related to sustainable entrepreneurial success.*

H5b. *Entrepreneurial resource acquisition is positively related to sustainable entrepreneurial success.*

3.4. Psychological Capital and Sustainable Entrepreneurial Success

Psychological capital implies positive psychological states, such as optimistic, resilient, and motivated, which significantly impact entrepreneurial success. Individuals with high levels of psychological capital will strive for their goals. Research shows that positive beliefs are significantly and positively correlated with performance, as demonstrated by work content outcomes [73], and such individuals can cope more optimistically with negative events, attribute positive events to permanent attributes, and promote enhanced job performance [74]. In addition, individuals with resilient psychological capital can adapt and overcome negative situations in entrepreneurial activities, leading to them achieving their desired outcomes [75]. Therefore, based on the above analysis, we propose the following hypotheses:

H6a. *Interpersonal psychological capital is positively related to sustainable entrepreneurial success.*

H6b. *Transactional psychological capital is positively related to sustainable entrepreneurial success.*

Figure 1 illustrates the theoretical framework of this paper.

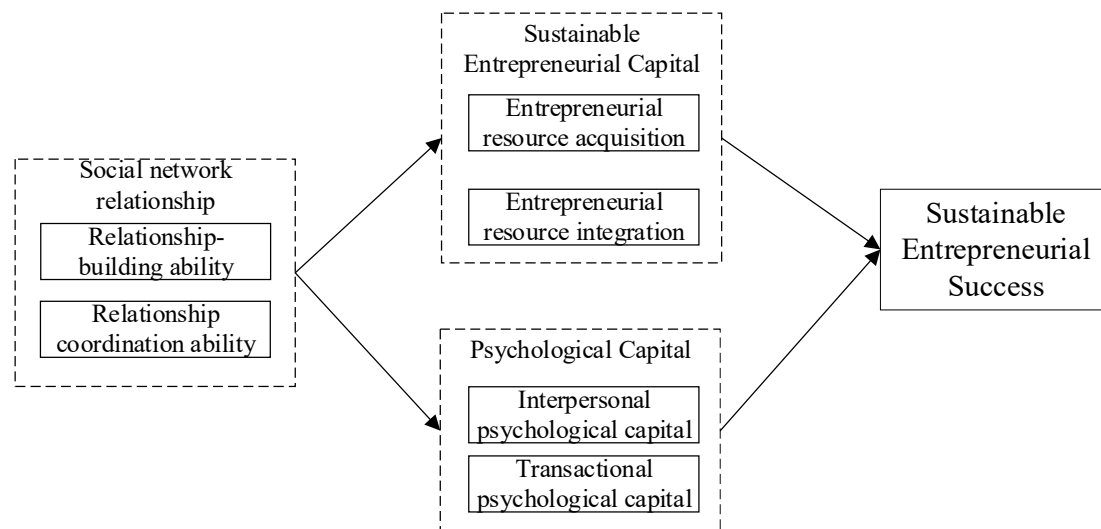


Figure 1. Theoretical framework.

4. Research Methodology

4.1. Measures

This study used measurement items validated in the existing literature to measure all the variables in the model. It used a Likert seven-point scale (in which a score of one represents strong disagreement and a score of seven represents strong agreement) to measure all the items. To ensure the validity of the measurement items, appropriate modifications were made to fit the research scenario of this study. For example, in the measurement of online relationship coordination ability, we adjusted “In our company we discuss regularly with our partners how we can support each other” in the original scale to “I almost always solve problems constructively with my partners through social media”. Measurement items suitable for this study are also added. (I can be flexible in dealing with my partners through social media). Table 1 shows the measurement items for each variable in detail.

Table 1. Measurement items and references.

Variable	Measurement Items	Reference
Online relationship-building ability (ORBA)	ORBA1: I have more opportunities to find new partners through social media. ORBA2: Through social media, I can analyze which potential companies want to achieve goals with me. ORBA3: Through social media, I can find new partners. ORBA4: I can build relationships with each partner based on their characteristics through social media.	[76]
Online relationship coordination ability (ORCA)	ORCA1: I can build good personal relationships with different business partners through social media. ORCA2: I can be flexible in dealing with my partners through social media. ORCA3: I almost always solve problems constructively with my partners through social media. ORCA4: I can coordinate with different partners through social media to accomplish my goals.	
Entrepreneurial resource integration (ERI)	ERI1: I will regularly source external technical knowledge for my use. ERI2: I regularly look for external ideas that may create value for me. ERI3: I have a well-established system for searching for and acquiring external technology and intellectual property. ERI4: I proactively contact external parties for better technical knowledge or products. ERI5: I tend to connect more with external parties and rely on their innovations. ERI6: I proactively buy external knowledge. ERI7: I make it a formal practice to buy and sell technical knowledge and intellectual property in the marketplace. ERI8: I have specialized units (e.g., caretakers, sponsors) to commercialize intellectual assets (e.g., sales, cross-licensing patents, business spin-offs). ERI9: I welcome others to buy and use my technical knowledge or intellectual property. ERI10: My core knowledge is sometimes unconsciously transferred to partners through unscheduled exchanges and interactions. ERI12: My proprietary knowledge is involuntarily transferred to partners without our permission.	[77]

Table 1. Cont.

Variable	Measurement Items	Reference
Entrepreneurial resource acquisition (ERA)	ERA1: I can obtain required funds from my friends and relatives. ERA2: It is not difficult for me to obtain funds from venture capital companies or banks. ERA3: I can obtain information about financial support policies from government departments. ERA4: My friends and relatives often recommend people with all kinds of talents. ERA5: I cannot quickly obtain funds from venture capital companies or banks. ERA6: I can obtain the talent and information I need from government departments. ERA7: Friends and relatives often recommend client resources. ERA8: I can obtain customer resources and information from various research institutes, organizations, and other sources. ERA9: I can obtain information about customer resources or market needs from the government.	[78]
Interpersonal psychological capital (IPC)	IPC1: I will be tolerant and forgiving regarding interpersonal relationships at work. IPC2: I will be humble and honest in my approach to interpersonal relationships at work. IPC3: I will be grateful and devoted to relationships at work. IPC4: I will treat interpersonal relationships at work with respect and courtesy.	[79]
Transactional psychological capital (TPC)	TPC1: I will be enterprising in matters of work. TPC2: I will be challenging and tenacious in my work. TPC3: I will be optimistic in my work. TPC4: I will be confident and brave in my work.	
Sustainable entrepreneurial success (SES)	SES1: The results of our company's sustainable business are satisfactory. SES2: The company has been very successful in implementing CSR. SES3: The firm has fully met our expectations (focus on future social and environmental impacts). SES4: With the help of social media, we have implemented the sustainability strategy. SES5: We have fulfilled our social mission (economic sustainability). SES6: We have fulfilled our social objectives (sustainable benefits of social development).	[80]

4.2. Sample Selection and Data Collection

To verify the validity of the questionnaire, this study was distributed to entrepreneurs with entrepreneurial experience. Before the formal distribution, we conducted a small-scale pre-survey, which collected 60 valid questionnaires. Based on the pre-survey data analysis results, we made appropriate adjustments to the questionnaire to improve the overall questionnaire's reliability and validity.

The formal investigation was conducted from May to July 2023. Based on the research theme and purpose of this paper, the method of online distribution is adopted. We distributed it through the questionnaire star platform. Respondents must complete all questions to be able to submit the questionnaire, so there are no missing values. We distributed the questionnaires to business managers and entrepreneurs in the Anhui Province in China. The group was made up of continuous entrepreneurs in various industries. A total of 361 questionnaires were collected, and 346 valid questionnaires were obtained after excluding unqualified questionnaires, such as too consistent answers and too fast response time, with an effective rate of 95.85%. The statistical characteristics of the sample are shown in detail in Table 2.

Table 2. Sample characterization (N = 346).

Items	Categories	Number	Percentage
Gender	Male	248	71.68%
	Female	98	28.32%
Age	18–25	12	3.47%
	26–30	26	7.51%
	31–40	143	41.33%
	41–50	131	37.86%
	51–60	32	9.25%
	>60	2	0.58%

Table 2. Cont.

Items	Categories	Number	Percentage
Industry	IT/Hardware and Software Services/E-commerce/Internet Operations	30	8.67%
	FMCG (Food/Beverage/Cosmetics)	10	2.89%
	Wholesale/Retail	26	7.51%
	Clothing/Textile/Leather	23	6.65%
	Furniture/Crafts/Toys	6	1.73%
	Education/Training/Research/Institutions	26	7.51%
	Home Appliances	1	0.29%
	Communications/Telecom Operations/Network Equipment/Value Added Services	3	0.87%
	Manufacturing	38	10.98%
	Automobile and Spare Parts	13	3.76%
	Catering/Entertainment/Tourism/Hotel/Life Services	11	3.18%
	Office Supplies and Equipment	1	0.29%
	Accounting/Auditing	3	0.87%
	Legal	3	0.87%
	Banking/Insurance/Securities/Investment Banking/Venture Funds	11	3.18%
	Electronic Technology/Semiconductor/Integrated Circuit	2	0.58%
	Instrumentation/Industrial Automation	3	0.87%
	Trade/Import and Export	4	1.16%
	Machinery/Equipment/Heavy Industry	3	0.87%
	Pharmaceutical/Bioengineering/Medical Equipment/Instrumentation	5	1.45%
	Medical/Nursing/Health Care/Sanitation	2	0.58%
	Advertising/PR/Media/Art	15	4.34%
	Publishing/Printing/Packaging	3	0.87%
	Real Estate Development/Construction Engineering/Decoration/Design	25	7.23%
	Property Management/Commercial Center	3	0.87%
	Intermediary/Consulting/Headhunting/Certification	18	5.20%
	Transportation/Logistics	7	2.02%
	Aerospace/Aviation/Energy/Chemical	3	0.87%
	Agriculture/Fishery/Forestry	16	4.62%
	Other Industries	32	9.25%
Education	Elementary and below	2	0.58%
	Junior high school	11	3.18%
	High school or junior college	47	13.58%
	University	186	53.76%
	Master's or above	100	28.90%

5. Data Analysis and Results

5.1. Measurement Model

To validate the reliability of the model, this study examined it in two main steps. First, this study used Cronbach's α (threshold of 0.70) to measure reliability. Next, this study assessed convergent validity using composite reliability (CR, with a threshold of 0.7) and average variance extracted (AVE, with a threshold of 0.5), combined with factor loadings (with a threshold of 0.6) [81]. The results of the specific data analysis are shown in detail in Table 3. Table 3 shows that the Cronbach's α values for the model variables in this study ranged from 0.893 to 0.967, which exceeds the threshold of 0.7, indicating that the questionnaire has good reliability [82]. The CR values ranged from 0.959 to 0.976, which exceeds the threshold of 0.7, indicating an excellent internal consistency of the variables in the model. The AVE values ranged from 0.718 to 0.909, exceeding the threshold of 0.5, indicating that the model's observed variables could explain each measurement dimension well [83]. These results indicate that the model has good convergent validity.

Table 3. Results of reliability and validity tests.

Variable	Factor Loadings	Cronbach's α	CR	AVE
TPC	0.795/0.88/0.9/0.902	0.893	0.926	0.757
IPC	0.947/0.957/0.953/0.956	0.967	0.976	0.909
SES	0.873/0.89/0.896/0.907/0.894/0.888	0.948	0.959	0.794
ERI	0.821/0.876/0.888/0.865/0.904/0.816/ 0.687/0.896/0.865/0.862/0.818	0.960	0.965	0.718
ERA	0.859/0.811/0.878/0.84/0.872/0.851/ 0.863/0.884/0.877	0.956	0.962	0.739
ORCA	0.935/0.937/0.928/0.952	0.954	0.967	0.880
ORBA	0.928/0.939/0.95/0.942	0.956	0.968	0.883

Note: ORBA = online relationship-building ability; ORCA = online relationship coordination ability; ERI = entrepreneurial resource integration; ERA = entrepreneurial resource acquisition; IPC = interpersonal psychological capital; TPC = transactional psychological capital; SES = sustainable entrepreneurial success.

In addition, the square roots of the AVE values were calculated to discriminate between the different validities of the variables, as shown in Table 4. The results show that the diagonal elements correspond to the square roots of the AVE values for each structure, and they are all greater than the values of the correlation coefficients corresponding to the other structures, which implies that a high degree of differentiation exists between the variables in the model [83].

Table 4. Results of the discriminant validity test.

Variable	TPC	IPC	SES	ERI	ERA	ORCA	ORBA
TPC	0.870						
IPC	0.505	0.953					
SES	0.443	0.609	0.891				
ERI	0.481	0.732	0.763	0.847			
ERA	0.531	0.838	0.697	0.810	0.860		
ORCA	0.747	0.580	0.470	0.567	0.611	0.938	
ORBA	0.767	0.549	0.491	0.537	0.541	0.883	0.940

Note: Bold numbers on the diagonal line are the square roots of AVE.

In addition, the heterogeneity–monotropy correlation ratio (HTMT) is another way to assess discriminant validity [84]. When the HTMT value is less than 0.85, it indicates good discriminant validity of the model. In Table 5, all HTMT values are less than 0.85.

Table 5. Heterogeneity–monotropy correlation ratio (HTMT).

Variable	TPC	IPC	SES	ERI	ERA	ORCA	ORBA
TPC							
IPC	0.544						
SES	0.478	0.634					
ERI	0.516	0.75	0.796				
ERA	0.576	0.841	0.73	0.836			
ORCA	0.803	0.603	0.493	0.59	0.639		
ORBA	0.823	0.571	0.516	0.558	0.566	0.825	

Note: ORBA = online relationship-building ability; ORCA = online relationship coordination ability; ERI = entrepreneurial resource integration; ERA = entrepreneurial resource acquisition; IPC = interpersonal psychological capital; TPC = transactional psychological capital; SES = sustainable entrepreneurial success.

5.2. Structural Model

To determine the validity of the overall measurement model, multiple indicators were reported. First, the VIF values for each measurement item were less than the recommended value of 10, indicating that the model did not suffer from multicollinearity. Second, the SRMR value of the model was 0.050, which is less than the threshold value of 0.08 [85], indicating that the model had a good fit.

Further results were tested by adding control variables. The results of the analysis showed that gender, age, literacy, and industry of engagement did not have a significant effect on the model; basic demographic information had no significant effect on the results. In addition, to assess the model's predictive ability, this study validated the value of Q^2 . When the value of Q^2 is greater than 0, it shows that the prediction of the partial least squares structural equation modeling (PLS-SEM) is less than the evaluation measurement error [86]. The analysis showed that the values of Q^2 for this model were all greater than 0, indicating good predictive performance of the model.

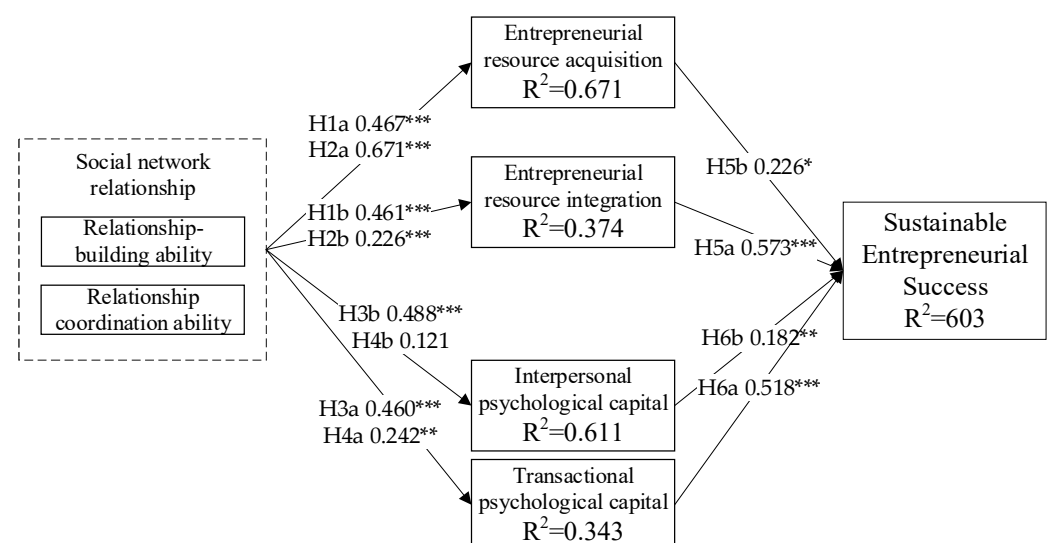
The validity of the model can also be affected if the model suffers from common method bias. Although the questionnaires for this study were distributed and collected in various ways, the sample was drawn from a wide range of sources, and the possibility of common method bias is low. However, this study still used Harman's one-factor test to identify potential common method bias [87]. Using principal component factor analysis, the first five factors collectively accounted for 75.33% of the total variance, and the first (largest) factor accounted for 33.44%, with no factor higher than 50%. This suggests a low likelihood of common method bias in this study.

Table 6 and Figure 2 shows the results, which are discussed in the following section.

Table 6. Results of hypothesis testing.

Hypothesis	Hypothetical Path	β	T Value	p Value	Conclusion
H1a	ORBA $\rightarrow$ ERI	0.467	4.64	0	Supported
H1b	ORBA $\rightarrow$ ERA	0.461	5.006	0	Supported
H2a	ORCA $\rightarrow$ ERI	0.671	14.63	0	Supported
H2b	ORCA $\rightarrow$ ERA	0.266	3.741	0	Supported
H3a	ORBA $\rightarrow$ TPC	0.460	4.918	0	Supported
H3b	ORBA $\rightarrow$ IPC	0.488	5.239	0	Supported
H4a	ORCA $\rightarrow$ TPC	0.242	2.493	0.013	Supported
H4b	ORCA $\rightarrow$ TPC	0.121	1.217	0.224	Not Supported
H5a	ERI $\rightarrow$ SES	0.573	5.668	0	Supported
H5b	ERA $\rightarrow$ SES	0.226	2.279	0.023	Supported
H6a	TPC $\rightarrow$ SES	0.518	8.046	0	Supported
H6b	IPC $\rightarrow$ SES	0.182	2.847	0.005	Supported

Note: ORBA = online relationship-building ability; ORCA = online relationship coordination ability; ERI = entrepreneurial resource integration; ERA = entrepreneurial resource acquisition; IPC = interpersonal psychological capital; TPC = transactional psychological capital; SES = sustainable entrepreneurial success.



* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Figure 2. Research result.

6. Discussion and Implications

6.1. Main Findings

This study aimed to reveal the mechanisms by which entrepreneurs' online social network relationships affect their sustainable entrepreneurial capital and sustainable entrepreneurial success. Based on our research objectives, by constructing a research model, collecting data, and testing hypotheses, we reached the following conclusions.

First, for sustainable entrepreneurs, online social network relationships significantly impact sustainable entrepreneurial capital. In particular, online relationship-building capacity has similar effects on sustainable entrepreneurial resource integration (H1a, path coefficient = 0.467, $p < 0.001$) and sustainable entrepreneurial resource acquisition (H1b, path coefficient = 0.461, $p < 0.001$). The use of social media can provide more opportunities for entrepreneurs to access sustainable entrepreneurial resources and to integrate those resources to increase their usefulness [88]. By contrast, compared to online relationship-building ability, online relationship coordination ability, although it can positively affect the acquisition of sustainable entrepreneurial resources (H2b, path coefficient = 0.266, $p < 0.001$), more predominantly affects the integration of sustainable entrepreneurial resources (H2a, path coefficient = 0.671, $p < 0.001$). This is consistent with previous studies [54], which show that when sustainable entrepreneurs have effective network relationship coordination skills, they can facilitate the integration of sustainable entrepreneurial resources and generate benefits for the firm.

Second, in terms of entrepreneurs' psychological capital, online relationship-building ability positively affects both transactional (H3a, path coefficient = 0.460, $p < 0.001$) and interpersonal (H3b, path coefficient = 0.488, $p < 0.001$) psychological capital. This ability enhances both types of psychological capital of sustainable entrepreneurs to support their sustainable entrepreneurial success. This finding also validates previous research findings [89]. Meanwhile, while the ability to coordinate online relationships positively influences transactional psychological capital (H3a, path coefficient = 0.242, $p = 0.013$) and does so more than the ability to build online relationships, the related hypothesis for interpersonal psychological capital (H4b, path coefficient = 0.121, $p = 0.224$) was unfounded. We suggest that this phenomenon occurs because although the coordination of sustainable entrepreneurial resources involves dealing with different people, the perception of their interpersonal relationships is relatively weak compared to that obtained via online relationship building. Coordination involves the interoperability and synergy of resources, often in one or more events, that lacks the continuity of the relational process.

Third, through the use of social media, sustainable entrepreneurial resource integration can contribute more effectively to sustainable entrepreneurial success (H7a, path coefficient = 0.573, $p < 0.001$) than can sustainable entrepreneurial resource acquisition (H7b, path coefficient = 0.702, $p = 0.023$). However, previously, we saw the result that online social network relationships have a more significant impact on acquiring sustainable entrepreneurial resources than on integrating those resources. Resource acquisition is not the same as effective use of those resources. Entrepreneurs can organize different types of resources into those available or scarce for their sustainable entrepreneurship enterprise (i.e., resource consolidation). In this way, those resources can provide better support during the sustainable entrepreneurial process, increasing the possibility of success.

Finally, the results of the analysis of the effect of psychological capital on sustainable entrepreneurial success indicate that transactional psychological capital has a more significant impact on sustainable entrepreneurial success (H7a, path coefficient = 0.518, $p < 0.001$) than does interpersonal psychological capital (H7b, path coefficient = 0.182, $p = 0.005$). Interpersonal relationships play an important role in the process of entrepreneurship, with the effective communication and coordination of stakeholders supporting success. However, sustainable entrepreneurial activity is market- and consumer-oriented; products and services are needed to make it happen. Completing a job or making a valuable product can bring core competencies to an enterprise. "Getting things working right" is an important step of sustainable entrepreneurial success. This may be why transactional (vs. interper-

sonal) psychological capital has a more significant effect on sustainable entrepreneurial success. These findings play an essential role in understanding the psychological changes during an individual's sustainable entrepreneurial process and in improving sustainable entrepreneurial success.

6.2. Implications

This study makes the following contributions. First, the results show the path mechanisms of the impact of online social network relationships on sustainable entrepreneurial success. The study advances the literature and explores the perspective of online social network relationship ability. Under the mechanisms explored, different dimensions of these relationships affect sustainable entrepreneurial capital and psychological capital, ultimately supporting sustainable entrepreneurial success. This valuable finding provides new empirical evidence of the role of networks formed in online social media environments. It also provides theoretical support to guide sustainable entrepreneurs to improve their abilities in terms of social network relationships.

Second, this study identifies the ways that different types of sustainable entrepreneurial capital influence entrepreneurial success. Specifically, resources acquired through online social network relationships must be integrated to strongly support success. This result, exploring how resources are used, is a helpful addition to the existing literature. It enriches and expands related theories of sustainable entrepreneurial capital and provides a new perspective for an in-depth study of how sustainable entrepreneurial resources are used. At the same time, the results of the study also provide theoretical guidelines to help entrepreneurs more efficiently use sustainable entrepreneurial resources and improve their methods of use.

Finally, the findings show the critical role of psychological capital in sustainable entrepreneurial success. Our work shows that transactional psychological capital has a more significant impact on this success than does interpersonal psychological capital. Although both are important in the process of sustainable entrepreneurship, improving the ability of an enterprise to deal with events that arise more strongly affects an entrepreneur's success. These findings echo existing findings while laying the groundwork for future research on psychological capital and sustainable entrepreneurial success. They also provide a theoretical basis and practical direction for sustainable entrepreneurs to take to improve their confidence and ability to succeed.

6.3. Limitations and Future Research

This study has the following shortcomings: First, this study focused on the sustainable entrepreneurial population in Anhui Province, China, so the diversity of the sample was limited. Future research should examine the mechanisms by which the characteristics of sustainable entrepreneurs from different countries and cultures, and how they use social media, affect sustainable entrepreneurial capital. In addition, this study is a cross-sectional study, which makes it challenging to examine the trends of competencies and sustainable entrepreneurial capital changes over time. Therefore, a time-series study will be considered to explore such changes. Finally, this study used questionnaires to collect data, so it lacked further validation and research. Therefore, multiple data collection methods will be considered in the future to increase the value and validity of the study.

7. Conclusions

Based on social network and sustainable entrepreneurship theories, this study explored how online social networks influence sustainable entrepreneurial success from the perspective of sustainable entrepreneurial capital. This study found that for entrepreneurs, online social network relationships significantly impact sustainable entrepreneurial capital. More specifically, online relationship-building ability positively affects transactional and interpersonal psychological capital. Through social media, sustainable entrepreneurial resource integration can contribute more effectively to entrepreneurial success than can

entrepreneurially resource acquisition. Transactional psychological capital has a more significant impact on sustainable entrepreneurial success than does interpersonal psychological capital.

The results confirm that online social networks affect sustainable entrepreneurial success by influencing sustainable entrepreneurial capital, indicating that the effective use of such networks can promote sustainable entrepreneurial success in the social media generation. The study results expand theories related to social networks, sustainable entrepreneurial capital, and psychological capital and provide practical support for entrepreneurs to promote and enhance their sustainable entrepreneurial success.

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